

# Relation Algebra

## §10. Freyd's Power allegories

(10a)

A **power allegory** is a division allegory with one unary operation on arrows,  $\exists$  **epsilon**, subject to

- 1  $\exists_R \square = R\square, \quad \exists_R = \exists_R \square$
- 2  $\mathbb{1} \sqsubseteq (R/\exists_R)(\exists_R/R) \quad \exists_R \text{ is **thick**}$
- 3  $\frac{\exists_R}{\exists_R} = \mathbb{1} \quad \exists_R \text{ is **straight**}$

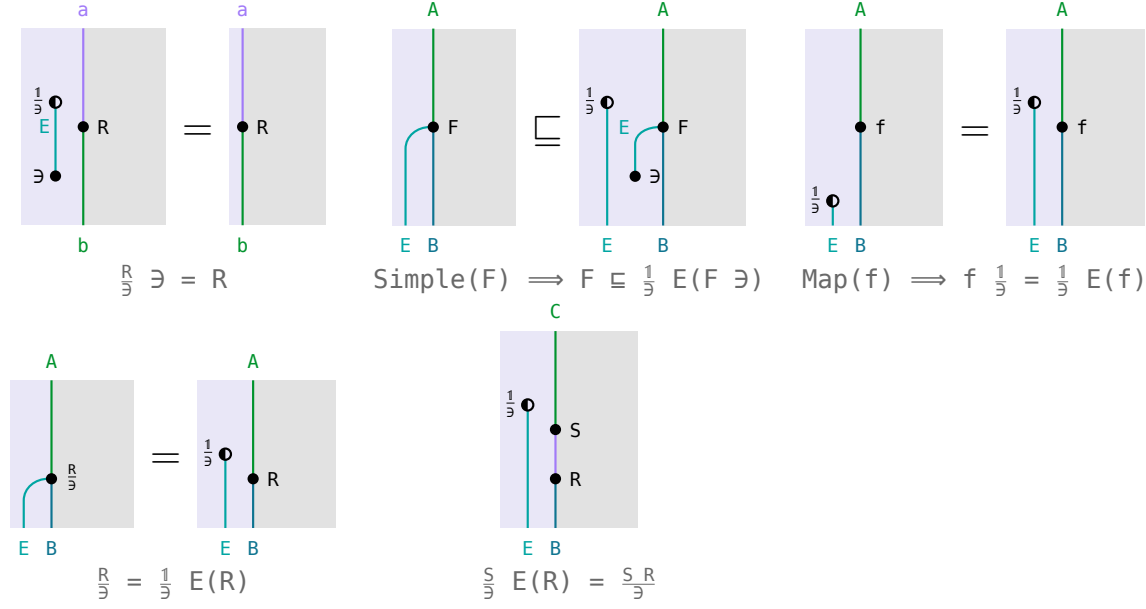
$R\square$  is  $R$ 's target, an identity arrow. For  $R : A \rightarrow B$  write  $\exists : PB \rightarrow B$ , dropping the subscript.

$\epsilon \triangleq \exists^\circ : A \rightarrow PA$

(10b)

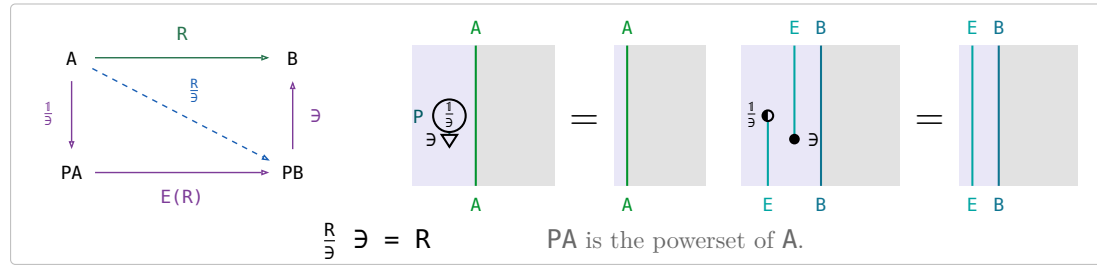
|    | law                                                                                                                                                                 | the reading                                                                                                                                                                                   |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | $\exists_R \square = R\square, \quad \exists_R = \exists_R \square$                                                                                                 | One $\exists$ per object, not per arrow.                                                                                                                                                      |
| 2  | $\exists$ is <b>thick</b>                                                                                                                                           | <b>Comprehension:</b> every $x$ has a set of exactly the people $x$ admires.                                                                                                                  |
| 3  | $\frac{R}{\exists} : A \rightarrow PB$ , for $R : A \rightarrow B$                                                                                                  | convert a relation to a function. $a \frac{R}{\exists} = \{b \mid a R b\}$                                                                                                                    |
| 4  | $\text{Map}(\frac{R}{\exists})$                                                                                                                                     |                                                                                                                                                                                               |
| 5  | $\frac{R}{\exists} \exists = R$                                                                                                                                     | reading the chosen set back through $\exists$ returns the relation.                                                                                                                           |
| 6  | $\text{Simple}(F) \implies F \sqsubseteq \frac{F \exists}{\exists}$                                                                                                 | A partial choice of sets is inside the total one.                                                                                                                                             |
| 7  | the <b>singleton map</b> is monic: $\frac{1}{\exists} (\frac{1}{\exists})^\circ \sqsubseteq \mathbb{1}$ two points with the same one-person set are the same point. | The one-person set.                                                                                                                                                                           |
| 8  | $\frac{\exists}{\exists} = \mathbb{1}$                                                                                                                              | Make the set of a set, then read it back one level down.                                                                                                                                      |
| 9  | <b>fusion:</b> $\text{Map}(f) \implies \frac{f R}{\exists} = f \frac{R}{\exists}$                                                                                   | Naturality of the unit, $\text{Map}(f) \implies f \frac{1}{\exists} = \frac{1}{\exists} E(f)$ . renaming a point and then taking its one-person set is taking the set and renaming inside it. |
| 10 | $\text{Map}(f) \wedge \mathbb{1}\square = \exists\square \implies \frac{f}{\exists} = f \frac{1}{\exists}$                                                          | Rename first or take singletons first — the fusion row above at $R=1$ .                                                                                                                       |
| 11 | $R\square = \exists\square \implies R /_s S = \frac{R}{\exists} (\frac{S}{\exists})^\circ$                                                                          | $x$ and $y$ match when $R$ sends $x$ and $S$ sends $y$ to the same set.                                                                                                                       |
| 12 | $E(R) \triangleq \frac{\exists R}{\exists}$                                                                                                                         | $E(R) : PA \rightarrow PB$ , image of a set of $A$                                                                                                                                            |
| 13 | $\frac{R}{\exists} = \frac{1}{\exists} E(R)$                                                                                                                        | $\frac{1}{\exists} : x \mapsto \{x\}$                                                                                                                                                         |
| 14 | $\frac{S}{\exists} E(R) = \frac{S R}{\exists}$                                                                                                                      | absorption — the monad's composition law, $\frac{S}{\exists} \diamond \frac{R}{\exists} = \frac{S R}{\exists}$ , §Section 10.3                                                                |
| 15 | $E(R) = E(\frac{R}{\exists})$ union                                                                                                                                 | the image of a set under $R$ is the union of the $R$ -sets of its points.                                                                                                                     |
| 16 | $\supseteq \triangleq \exists / \exists$                                                                                                                            | $xs \supseteq ys \iff \forall a. ys \exists a \rightarrow xs \exists a$                                                                                                                       |
| 17 | $\subseteq = \supseteq^\circ$                                                                                                                                       | $xs \subseteq ys \iff ys \supseteq xs$                                                                                                                                                        |
| 18 | $T^\circ \frac{T}{\exists} \sqsubseteq \epsilon$                                                                                                                    | every $T$ -value of a point is a member of its $T$ -set: $\wedge(T)$ is a map and $\wedge(T)\exists = T$ .                                                                                    |

(10c)



### §10.1. $\mathbf{i}\text{-E}$ Power Allegory defined as adjunction

(10.1a)



$\mathbf{i}$  is the inclusion  $\text{Map}(\mathcal{A}) \rightarrow \mathcal{A}$ , doing nothing, so  $E$  is both the functor and the monad  $\mathbf{i}E$ .

### §10.2. $E \setminus$ as a composite

| $E \setminus Z = \subseteq (\frac{Z^\circ}{3})^\circ$                                                                         |                                               |                                                                      | (10.2a) |
|-------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|----------------------------------------------------------------------|---------|
| $xs$ is related to $c$ by $E \setminus Z$ exactly when $xs \subseteq \Lambda(Z^\circ)(c)$ , the set of every $a$ with $a Z c$ |                                               |                                                                      |         |
|                                                                                                                               |                                               |                                                                      |         |
|                                                                                                                               |                                               |                                                                      |         |
|                                                                                                                               |                                               |                                                                      |         |
| (a)                                                                                                                           | $E \setminus Z$                               | $Z = E \Lambda(Z^\circ)^\circ \text{ — } \frac{R}{3} \exists = R$    |         |
| (b)                                                                                                                           | $= E \setminus (E (\frac{Z^\circ}{3})^\circ)$ | $X \setminus (Y f^\circ) = (X \setminus Y) f^\circ, f \text{ a map}$ |         |
| (c)                                                                                                                           | $= \subseteq (\frac{Z^\circ}{3})^\circ$       |                                                                      |         |

| $R \setminus S = \frac{R^\circ}{3} \subseteq (\frac{S^\circ}{3})^\circ$                                                                                              |                                                           |                                                                                                                                             | (10.2b) |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|---------|
| $b$ is related to $c$ by $R \setminus S$ exactly when $\Lambda(R^\circ)(b) \subseteq \Lambda(S^\circ)(c)$ , the $R$ -preimage of $b$ inside the $S$ -preimage of $c$ |                                                           |                                                                                                                                             |         |
|                                                                                                                                                                      |                                                           |                                                                                                                                             |         |
|                                                                                                                                                                      |                                                           |                                                                                                                                             |         |
|                                                                                                                                                                      |                                                           |                                                                                                                                             |         |
|                                                                                                                                                                      |                                                           |                                                                                                                                             |         |
| (a)                                                                                                                                                                  | $R \setminus S$                                           | $R = E \Lambda(R^\circ)^\circ \text{ — } \frac{R}{3} \exists = R$                                                                           |         |
| (b)                                                                                                                                                                  | $= (E (\frac{R^\circ}{3})^\circ) \setminus S$             | $(XY) \setminus S = Y \setminus (X \setminus S)$                                                                                            |         |
| (c)                                                                                                                                                                  | $= (\frac{R^\circ}{3})^\circ \setminus (E \setminus S)$   | $f^\circ \setminus X = fX; E \setminus S = \subseteq \Lambda(S^\circ)^\circ \text{ — } E \setminus Z = \subseteq (\frac{Z^\circ}{3})^\circ$ |         |
| (d)                                                                                                                                                                  | $= \frac{R^\circ}{3} \subseteq (\frac{S^\circ}{3})^\circ$ |                                                                                                                                             |         |

§10.3.  $\mathcal{A} \approx \text{Kleisli}(E)$ 

(10.3a)

$$E(R) \triangleq \frac{\exists}{\exists} R, \quad \frac{1}{\exists} : A \rightarrow PA, \quad \text{union} = E(\exists) : PPA \rightarrow PA$$

$$f \diamond g \triangleq f \circ E(g) \circ \text{union}, \quad \text{for } f : A \rightarrow PB \text{ and } g : B \rightarrow PC$$

the monad is on  $\text{Map}(\mathcal{A})$ , not on the allegory:  $E$  is a relator on all relations, but  $\frac{1}{\exists}$ ,  $\text{union}$  and  $f \circ E(g) \circ \text{union}$  are maps, and the Kleisli construction happens where they live.

(10.3b)

$$\begin{aligned} \boxed{\frac{S}{\exists} \diamond \frac{R}{\exists}} & \xrightarrow{\text{definition of } \diamond, \text{ union}=E(\exists)} \boxed{\frac{S}{\exists} E\left(\frac{R}{\exists}\right) E(\exists)} \xrightarrow{E \text{ a functor, } E(X)E(Y)=E(XY)} \boxed{\frac{S}{\exists} E\left(\frac{R}{\exists}\exists\right)} \\ & \xrightarrow{\frac{R}{\exists} \exists = R, \frac{R}{\exists} \exists = R} \boxed{\frac{S}{\exists} E(R)} \xrightarrow{\frac{S}{\exists} E(R) = \frac{S}{\exists} R, \text{ absorption}} \boxed{\frac{SR}{\exists}} \end{aligned}$$

the first three steps only unfold  $\diamond$  —  $E$  is a functor, so  $E\left(\frac{R}{\exists}\right)\text{union}=E\left(\frac{R}{\exists}\exists\right)=E(R)$  and the  $\text{union}$  is gone.

Absorption,  $\frac{S}{\exists} E(R) = \frac{S}{\exists} R$ , is the whole law, and it is the functoriality of  $\frac{\cdot}{\exists}$ .

$1$  goes to  $\frac{1}{\exists}$ , the Kleisli identity, by definition — so  $\frac{\cdot}{\exists}$  is an isomorphism of categories  $\mathcal{A} \approx \text{Kleisli}(E)$ , and §Section 10.1's  $i \dashv E$  is the Kleisli adjunction.  $i$  is the identity on objects, so every object of the allegory is free.